

Ying Kung

Personal Website: <https://phd-portfolio-jw42.vercel.app/> | Tel: (+886)976976829 | Email: evan08292001@gmail.com

EDUCATION

National Taiwan University (NTU)

M.S. in applied mechanics

GPA: 3.67/4.3

Relevant courses: Applications of Deep Learning in Science and Engineering; Principles and Experiments of Mechatronic Systems; Stochastic Control; Electronics Laboratory

National Yang Ming Chiao Tung University (NYCU)

B.S. in Civil Engineering

Taiwan

Aug 2023 — Jul 2025

Taiwan

Aug 2019 — Jul 2023

RESEARCH INTERESTS

Robotics, Control and Estimation, Motion Planning, Intelligent Mechatronic Systems, Autonomous and Electric Vehicles, Battery Diagnostics and System Health Monitoring.

PUBLICATIONS

- **Ying Kung**, Yu-Hong Zhang, Chi-Jyun Ko, & Kuo-Ching Chen (2024). Predicting the onset of lithium plating by the full-cell voltage: A pseudo-P curve approach. Under review, *Energy Storage Materials*.

CONFERENCE PRESENTATIONS

- **Ying Kung**. Onset prediction of lithium plating from full-cell voltage and its application to charging strategies. Presented at the *35th Annual Conference on Combustion and Energy*, National Tsing Hua University, Hsinchu, Taiwan, May 2025.
- **Ying Kung**. Frequency-specific modeling for battery parameter identification under battery aging. Accepted for oral presentation at *The 2025 International Conference on Green Electrochemical Technologies & 2025 Annual Meeting of Electrochemical Society of Taiwan (ICGET-Tw)*, National Cheng Kung University, Tainan, Taiwan, scheduled for Oct. 31 – Nov. 2, 2025.

RESEARCH PROJECTS

Research on Lithium Plating Onset Prediction and Charging Strategy Optimization

Graduate Research | Advisor: Prof. Kuo-Ching Chen

NTUIAM, Nov 2024 — Apr 2025

- Developed a pseudo-P curve method to detect lithium plating onset in LIBs from full-cell voltage only, eliminating the need for reference electrodes.
- Validated the method via PyBaMM simulations at 0 °C, achieving onset prediction errors within 3.44% of total capacity.
- Designed multi-stage constant-current step-charging protocols that completely prevent lithium plating, reaching 99.72% Coulombic efficiency at 0 °C.
- Evaluated charging strategies across various C-rates and initial SOCs, providing guidelines for safe and efficient operation.

Frequency-Specific Modeling for Battery Parameter Identification Under Aging

Graduate Research | Advisor: Prof. Kuo-Ching Chen

NTUIAM, May 2025 — Present

- Developed a physics-informed, segment-specific ML framework for estimating electrochemical parameters from partial Electrochemical Impedance Spectrum (EIS) measurements without current-voltage history.
- Applied two-stage Sobol sensitivity analysis to identify the most sensitive parameters in each frequency band, enabling targeted regression.
- Trained compact Transformer models on 20,000 simulated samples per band, achieving < 1% MAPE for SOH $\geq$ 75% and < 3% under severe aging, with inference time of $\approx$ 0.3s.
- Reduced computation cost by 5 orders of magnitude compared to full-spectrum GA inversion, enabling real-time onboard BMS parameter identification using 9–19 frequency points.

Concrete Boat Design and Fabrication Project

Undergraduate Capstone Project | Advisor: Assoc. Prof. Hsin-yu Shan

NYCUCV, Jan 2023 — Jun 2023

- Designed and built a functional concrete boat inspired by international collegiate competitions.
- Performed buoyancy and structural calculations; created CAD models in Rhino and Fusion 360.
- Produced molds via 3D printing and polystyrene cutting; developed custom concrete mix for high workability.

AWARDS & SCHOLARSHIPS

- Honorable Mention, Student Paper Competition, 35th National Conference on Combustion and Energy May 2025
- Dean's Award, College of Engineering, National Taiwan University Aug 2025

SKILLS & INTERESTS

- **Programming & Modeling:** Python, MATLAB, COMSOL
- **System Modeling & Analysis:** Dynamic systems, parameter estimation, data-driven modeling
- **Battery Systems:** PyBaMM, electrochemical modeling, EIS analysis
- **Documentation & Tools:** LaTeX
- **Interests:** Fitness, Gaming, Food Exploration